LESSON 01 / 04

Status of this sheet

The substitution $t = \tan\frac{A}{2}$ (Weierstrass) is **standard RMO/INMO technique** and a bridge to integral calculus; every model below is derivable from Part I identities. Nothing here "can never appear in JEE" — historically JEE-style papers have used it in disguised form; the classification is only that it exceeds the syllabus sentence "Trigonometric functions, general solution, identities".

The substitution and what it buys

Put $t = \tan\dfrac{A}{2}$. Then (for $A \neq \pi + 2k\pi$):

$$\sin A = \frac{2t}{1+t^2}, \qquad \cos A = \frac{1-t^2}{1+t^2}, \qquad \tan A = \frac{2t}{1-t^2}.$$

SHORT DERIVATION

Derivation — divide the double-angle forms by $\cos^2\frac{A}{2}$

$\sin A = 2\sin\frac{A}{2}\cos\frac{A}{2} = \frac{2\tan\frac{A}{2}}{\sec^2\frac{A}{2}} = \frac{2t}{1+t^2}$ (using $\sec^2 = 1+\tan^2$). Similarly $\cos A = \cos^2\frac{A}{2} - \sin^2\frac{A}{2} = \frac{1-t^2}{1+t^2}$ after dividing by $\cos^2\frac{A}{2}$. Tangent: ratio of the two ✓.

The point: **any rational expression in $\sin A, \cos A$ becomes a rational function of t** — trig equations become polynomial equations; trig integrals $\int R(\sin x, \cos x)\,dx$ become rational integrals (calculus toolkit).

WORKED EXAMPLE

Worked. Solve $\sin x + \cos x = 1$ for $0 \le x < 2\pi$ with $t = \tan\frac{x}{2}$

$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1 \Rightarrow 2t + 1 - t^2 = 1 + t^2 \Rightarrow 2t^2 - 2t = 0 \Rightarrow t = 0$ or $t = 1$. $t = 0 \Rightarrow x = 0$; $t = 1 \Rightarrow \frac{x}{2} = \frac{\pi}{4} \Rightarrow x = \frac{\pi}{2}$. The excluded point $x = \pi$ (where $\cos\frac{x}{2} = 0$) gives $\sin\pi + \cos\pi = -1 \neq 1$ ✓ nothing lost. Answer: $x \in \{0, \frac{\pi}{2}\}$ ✓.

COMMON TRAP

The blind spot — $A = \pi$ (mod 2π)

$t = \tan\frac{A}{2}$ is undefined when $\frac{A}{2} = \frac{\pi}{2} + k\pi$, i.e. $A = \pi + 2k\pi$. Every solution there is invisible to the substitution — **always test $A = \pi + 2k\pi$ in the original equation** and add it if it works. For $A \in (-\pi, \pi)$ the map $A \mapsto t$ is a bijection onto $\mathbb{R}$ (this is the clean domain window for the substitution).

KEY RESULT

When to use it. (1) Equations rational in $\sin, \cos$ that resist factorisation; (2) proving identities by clearing denominators in t; (3) converting geometry (slope $= \tan\frac{\theta}{2}$) into algebra; (4) (forward reference) any integral $\int R(\sin x, \cos x)\,dx$. Counter-indication: when half-angle or product-to-sum is quicker — do the cheap tool first.

Worked models

WORKED EXAMPLE

Worked. Solve $\sin x + \cos x = 1$ over all reals

General solutions from the two families: $x = 2n\pi$ (from $t = 0$) and $x = \frac{\pi}{2} + 2n\pi$ (from $t = 1$), $n \in \mathbb{Z}$; the blind-spot test $x = \pi + 2n\pi$ fails. Combined: $x = 2n\pi$ or $x = \frac{\pi}{2} + 2n\pi$ ✓ (leave as the two families).

WORKED EXAMPLE

Worked. Express $\frac{1-\cos x}{\sin x}$ in t

$\frac{1-\cos x}{\sin x} = \frac{1 - \frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}} = \frac{\frac{2t^2}{1+t^2}}{\frac{2t}{1+t^2}} = t = \tan\frac{x}{2}$ ✓ — consistent with the half-angle identity of Part I. Any rational $R(\sin x, \cos x)$ now reduces by t-clearing denominators; $1 + t^2 \neq 0$ is automatic for real t.

Assumptions & conventions

A real with $A \neq \pi + 2k\pi$ under the substitution; the blind spot is checked separately in every problem; rational expressions in $\sin, \cos$ mean ratios of polynomials; all conversions verified back in the original equation.

LESSON 02 / 04

Status of this sheet

*Products like $\prod \sin \frac{k\pi}{n}$ are **classic RMO/INMO algebra** obtained by combining Part I identities with complex numbers (chapter 5 of the main sequence supplies $z = re^{i\theta}$ and $|z|$; only elementary modulus facts are used). Classification: beyond the syllabus sentence, but every step is derivable from supplied identities and the definition of $|1 - e^{i\theta}|$.*

The two flagship products

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$$

$$\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = \frac{1}{2^n}$$

SHORT DERIVATION

Derivation — factor $z^n - 1$ at the roots of unity, set $z = 1$, take moduli

The n roots of $z^n = 1$ are $e^{2\pi ik/n}$, $k = 0, \dots, n-1$. So $z^n - 1 = (z-1)\prod_{k=1}^{n-1}(z - e^{2\pi ik/n})$.

Cancel $z - 1$ ($\frac{z^n-1}{z-1} = 1 + z + \cdots + z^{n-1}$) and set $z = 1$: $n = \prod_{k=1}^{n-1}(1 - e^{2\pi ik/n})$. Now $|1 - e^{i\theta}| = 2\left|\sin \frac{\theta}{2}\right|$ (chord of a unit-circle arc; expand $|1 - \cos\theta - i\sin\theta|$). Taking moduli: $n = \prod 2\sin \frac{k\pi}{n} = 2^{n-1}\prod \sin \frac{k\pi}{n}$ (each $\sin \frac{k\pi}{n} > 0$ for $1 \le k \le n-1$) ✓. The cosine product comes from the same factorisation at $z = -1$.

WORKED EXAMPLE

Worked. $\prod_{k=1}^{6} \sin \frac{k\pi}{7}$ and $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$

$n = 7$: product $= \frac{7}{2^6} = \frac{7}{64} = 0.109375$ ✓ (direct: $0.43388 \times 0.78183 \times 0.97493 \times 0.97493 \times 0.78183 \times 0.43388 = 0.109375$).

Cosine product with $2n + 1 = 7$ ($n = 3$): $\frac{1}{2^3} = \frac{1}{8}$ ✓ (direct: $0.90097 \times 0.62349 \times 0.22252 = 0.125$).

The pattern to internalise

A product of trig values at rational multiples of π is a roots-of-unity problem in disguise. *Second tool: when the angles come from a polynomial whose roots are $\tan$-values or the values themselves, **Vieta** hands you sums/products for free (see the $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$ model of Part I sheet 8 — same idea with $\tan 3\theta$).*

WORKED EXAMPLE

Worked. Sine product for $n = 5$: $\sin 36^\circ \sin 72^\circ$?

$\prod_{k=1}^{4} \sin \frac{k\pi}{5} = \frac{5}{16}$. *Pair* $\sin \frac{\pi}{5} = \sin 36^\circ$, $\sin \frac{2\pi}{5} = \sin 72^\circ$, $\sin \frac{3\pi}{5} = \sin 72^\circ$, $\sin \frac{4\pi}{5} = \sin 36^\circ$: *product* $= (\sin 36^\circ \sin 72^\circ)^2 = \frac{5}{16}$, *so* $\sin 36^\circ \sin 72^\circ = \frac{\sqrt{5}}{4} \approx 0.5590$ ✓ *(direct:* $0.58779 \times 0.95106 = 0.55902$*)*.

COMMON TRAP

Traps on this sheet

- $\prod_{k=1}^{n-1}$ *vs* $\prod_{k=1}^{n}$*: the sine product has* $n - 1$ *factors (the* $k = 0$ *factor would be* $\sin 0 = 0$*); count factors before quoting.*
- $|1 - e^{i\theta}| = 2|\sin \frac{\theta}{2}|$ *— drop the absolute values only after checking positivity (done via* $0 < \frac{k\pi}{n} < \pi$*).*
- *The cosine product* $\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = 2^{-n}$ *has* n *factors and denominator* $2n + 1$ *— a different index bookkeeping from the sine product.*
- *Claims here need complex arithmetic: state* $e^{i\theta} = \cos\theta + i\sin\theta$ *and* $|z|$ *facts explicitly in a solution write-up (assumptions below).*

Assumptions & conventions

Complex numbers used: $e^{i\theta} = \cos\theta + i\sin\theta$, $|zw| = |z||w|$, $|1 - e^{i\theta}| = 2|\sin \frac{\theta}{2}|$ *(proved above in modulus form). All products finite;* $n \geq 2$ *integer; positivity of* $\sin \frac{k\pi}{n}$ *for* $1 \leq k \leq n - 1$ *used to drop absolute values.*

LESSON 03 / 04

Status of this sheet

*The telescoping series $\sum \sin k\theta$ and the sums $\sum \cot^2$, $\sum \csc^2$ are **standard olympiad algebra**; all derive from Part I transformation formulae and the roots-of-Vieta technique of sheet 2. Models are derivable from supplied information; classified as extensions because they sit beyond the current syllabus sentence.*

Sums with arguments in arithmetic progression (telescoping)

The trick: *multiply by $2\sin\frac{\theta}{2}$ and watch the product-to-sum formula telescope.*

$$\sum_{k=1}^{n} \sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

$$\sum_{k=1}^{n} \cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

SHORT DERIVATION

Derivation — the $2\sin\frac{\theta}{2}$ multiplier

$2\sin\frac{\theta}{2}\sin k\theta = \cos\left(k-\frac{1}{2}\right)\theta - \cos\left(k+\frac{1}{2}\right)\theta$. *Summing over $k = 1 \ldots n$ telescopes to* $\cos\frac{\theta}{2} - \cos\left(n+\frac{1}{2}\right)\theta$. *Then apply* $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$ *with* $C = \frac{\theta}{2}$, $D = \left(n+\frac{1}{2}\right)\theta$; $= 2\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}$. *Divide by* $2\sin\frac{\theta}{2}$ ✓. *(Valid for* $\sin\frac{\theta}{2} \neq 0$, *i.e.* $\theta \notin 2\pi\mathbb{Z}$; *the excluded cases are computed directly as sums of 0/1.)*

WORKED EXAMPLE

Worked check. $n = 3$, $\theta = \frac{\pi}{6}$

Direct: $\sin 30^\circ + \sin 60^\circ + \sin 90^\circ = 0.5 + 0.86603 + 1 = 2.36603$. *Formula:* $\frac{\sin 45^\circ \sin 60^\circ}{\sin 15^\circ} = \frac{0.70711 \times 0.86603}{0.25882} = 2.36603$ ✓.

Sums of squares at rational multiples of π (roots + Vieta)

The numbers $t_k = \tan\frac{k\pi}{n}$, $k = 1, \ldots, n-1$, are the non-zero roots of a polynomial obtained from $\tan(n\theta)$ in terms of $t = \tan\theta$ (write $\tan n\theta = \frac{\Im(1+it)^n}{\Re(1+it)^n}$ and set the numerator to 0). Vieta then gives sums of powers.

$$\sum_{k=1}^{n-1} \cot^2\frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}, \qquad \sum_{k=1}^{n-1} \csc^2\frac{k\pi}{n} = \frac{n^2-1}{3}.$$

SHORT DERIVATION

Derivation sketch — cotangents as roots of the complementary polynomial

$\cot^2\frac{k\pi}{n} = \frac{1}{t_k^2}$, and the symmetric identity $\sum\frac{1}{t_k^2} = \left(\sum\frac{1}{t_k}\right)^2 - 2\sum_{j<k}\frac{1}{t_j t_k}$ turns Vieta's data for the t_k into the closed form $\frac{(n-1)(n-2)}{3}$. Then $\csc^2 = 1 + \cot^2$ adds $(n-1)$ terms of 1: $\sum\csc^2 = (n-1) + \frac{(n-1)(n-2)}{3} = \frac{(n-1)(n+1)}{3} = \frac{n^2-1}{3}$ ✓.

WORKED EXAMPLE

Worked checks. $n = 3, 4, 5$

$\cot^2$: $n=3$: $\cot^2 60^\circ + \cot^2 120^\circ = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} = \frac{2\cdot 1}{3}$ ✓. $n=4$: $1 + 0 + 1 = 2 = \frac{3\cdot 2}{3}$ ✓. $n=5$: $4 = \frac{4\cdot 3}{3}$ ✓. $\csc^2$: $n=3$: $\frac{4}{3} + \frac{4}{3} = \frac{8}{3} = \frac{9-1}{3}$ ✓; $n=4$: $2 + 1 + 2 = 5 = \frac{15}{3}$ ✓; $n=5$: $8 = \frac{24}{3}$ ✓.

KEY RESULT

Whenever a sum runs over trig values at equally spaced angles: (1) look for a telescoping multiplier ($2\sin\frac{\theta}{2}$ for AP-argument sums); (2) look for the polynomial whose roots they are (Vieta for square/power sums); (3) pair symmetric terms ($k \leftrightarrow n-k$) to halve the work; (4) validate a small case numerically before the proof.

COMMON TRAP

Traps on this sheet

- The telescoping formulas divide by $\sin\frac{\theta}{2}$: state $\theta \notin 2\pi\mathbb{Z}$ (or handle those cases directly).
- $\sum\sin k\theta$ closed form: the numerator is $\sin\frac{n\theta}{2}\cdot\sin\frac{(n+1)\theta}{2}$ — order $(n+1)$ vs n swapped between the two factors vs the cosine version (which has $\cos\frac{(n+1)\theta}{2}$).
- $\cot^2\frac{k\pi}{n}$ has a term at $k = \frac{n}{2}$ (when n even) equal to 0 — include it (it is $\cot^2\frac{\pi}{2} = 0$).
- $\tan n\theta$'s polynomial in $t = \tan\theta$ has **degree** n in the tangent-angle formula bookkeeping; only the roots t_k with $\tan$ defined count.

Assumptions & conventions

$n \geq 2$ integer; θ real with $\sin\frac{\theta}{2} \neq 0$ in series formulas; Vieta applied to real polynomials; numeric checks above computed in radians (converted to degrees for readability).

LESSON 04 / 04

Status of this sheet

Important classification note. The current JEE Advanced syllabus sentence for trigonometry is "Trigonometric functions. General solution of trigonometric equations. Identities." — it does **not** list properties of triangles or heights & distances. The formulae below are the **standard triangle toolkit** used across JEE Main-level papers, olympiads (RMO/INMO geometry-algebra), and later chapters (conics, 3D, vectors). They are added theory (not derivable from the identities alone — they encode triangle geometry), but every model built here only uses Part I identities plus the sine/cosine rules as stated. Do not assume they "can never appear"; assume they are simply outside the quoted syllabus sentence and learn them as toolkit.

The four triangle formulae

For a triangle with sides a, b, c opposite angles A, B, C, circumradius R, inradius r, semiperimeter $s = \frac{a+b+c}{2}$, area Δ:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A = \sqrt{s(s-a)(s-b)(s-c)}, \qquad r = \frac{\Delta}{s}$$

$$\tan\frac{A}{2} = \frac{r}{s-a}$$

SHORT DERIVATION

Cosine rule and $\Delta = rs$ — one line each

Cosine rule: drop a height $h = c\sin B$ from C; project: $a = b\cos C + c\cos B$-style coordinate computation gives $a^2 = b^2 + c^2 - 2bc\cos A$ (or expand $|\vec{B} - \vec{C}|^2$ in the vectors chapter). $\Delta = rs$: split the triangle into three (r, s)-shaped pieces — $\Delta = \frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc = rs$ ✓. $\tan\frac{A}{2} = \frac{r}{s-a}$: the incircle tangent lengths are $s-a, s-b, s-c$ and half-angle right triangles give the ratio.

Derived special values (pure identities, pentagon angles)

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.30902$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4} \approx 0.80902$$

SHORT DERIVATION

Derivation — double-angle, then quadratic

Let $x = 18^\circ$, so $5x = 90^\circ$ and $2x = 90^\circ - 3x$, hence $\sin 2x = \cos 3x$. Expand: $2\sin x\cos x = 4\cos^3 x - 3\cos x$; with $\cos x \neq 0$: $2\sin x = 4\cos^2 x - 3 = 4(1 - \sin^2 x) - 3 = 1 - 4\sin^2 x$. So $4\sin^2 x + 2\sin x - 1 = 0 \Rightarrow \sin x = \frac{-2+\sqrt{4+16}}{8} = \frac{\sqrt{5}-1}{4}$ (positive root; $\sin 18^\circ > 0$) ✓. Then $\cos 36^\circ = 1 - 2\sin^2 18^\circ = \frac{\sqrt{5}+1}{4}$ ✓ (check numerically: 0.809017).

WORKED EXAMPLE

Worked. Triangle with $A = 60^\circ$, $b = 5$, $c = 8$; find a and Δ

Cosine rule: $a^2 = 25 + 64 - 2 \cdot 5 \cdot 8 \cdot \frac{1}{2} = 49 \Rightarrow a = 7$. $\Delta = \frac{1}{2} \cdot 5 \cdot 8 \cdot \frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.32$. Cross-check with Heron: $s = \frac{7+5+8}{2} = 10$, $\sqrt{10 \cdot 3 \cdot 5 \cdot 2} = \sqrt{300} = 10\sqrt{3}$ ✓.

WORKED EXAMPLE

Worked. R and r for the same triangle

$2R = \frac{a}{\sin A} = \frac{7}{\sqrt{3}/2} = \frac{14}{\sqrt{3}} \Rightarrow R = \frac{7}{\sqrt{3}} \approx 4.041$. $r = \frac{\Delta}{s} = \frac{10\sqrt{3}}{10} = \sqrt{3} \approx 1.732$. Check $\tan\frac{A}{2} = \tan 30^\circ = \frac{1}{\sqrt{3}}$ vs $\frac{r}{s-a} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$ ✓.

COMMON TRAP

Traps — triangle toolkit

- The sine rule gives an **ambiguous case** (SSA): $a = b\frac{\sin A}{\sin B}$ with $\sin B = \frac{b\sin A}{a}$ may admit two triangles (B and $180^\circ - B$).
- $\Delta = \frac{1}{2}$(base)(height) needs the height; $\frac{1}{2}bc\sin A$ needs the angle **between** the two sides.
- Heron's formula: s is semiperimeter; writing $s = a + b + c$ is the classic slip.
- $\tan\frac{A}{2} = \frac{r}{s-a}$ requires the usual triangle convention ($s > a$); for degenerate/failing configurations state the inequality.
- 18°-family values ($\sin 18^\circ$, $\cos 36^\circ$, $\sin 54^\circ = \cos 36^\circ$ etc.): derive from the quadratic each time if memory fails — and pick the sign from the quadrant (18° is QI).

Assumptions & conventions

Triangle = non-degenerate (positive angles summing to π, positive sides satisfying triangle inequalities); R circumradius, r inradius, s semiperimeter, Δ area; special values derived from double-angle identities as shown; toolkit status per the note at the top of this sheet.